Topic 15: Carboxylic Acids and Derivatives

Unit XIV: naming, properties, acidity, reactions, polymers

CHEMISTRY (PROF. L. RANAY) | FINALS EVAL, NOV 14: TOPICS 15-18 (UNITS XIV-XVII)

Coverage

This file covers **Units XIV, XV, XVI and XVII** (topics 15-18). **Unit XVIII will be added later.**

Last chapter of organic chemistry

Four groups: **carboxylic acids, amides, esters, anhydrides**. For each: **naming, physical properties, reactions**. A carboxyl group is a carbonyl with an **OH** on the carbonyl carbon (an aldehyde has an H there). Carboxylic acids are **real (weak) acids**; amines are bases.

NAMING CARBOXYLIC ACIDS

- Longest chain holding the carboxyl group: **drop -e, add -oic acid** (methanoic, ethanoic, propanoic, butanoic, hexanoic). The COOH carbon is **always C1** (terminal): no locator.
- **Highest priority** group, so OH and NH_2 become **hydroxy-** and **amino-**: **5-hydroxyhexanoic acid**, **4-aminobenzoic acid (PABA)**. Numbering: functional group → double bond → nearest branch → alphabetical.
- **Dicarboxylic acids: -dioic acid, KEEP the -e**: ethanedioic (oxalic), propanedioic (malonic), butanedioic (succinic), pentanedioic (glutaric), hexanedioic (adipic). Seen in the **TCA / Krebs cycle**.
- **4-Aminobutanoic acid = γ-aminobutyric acid = GABA** (α, β, γ locate substituents).
- **Common names: formic** C1 (ant), **acetic** C2 (vinegar), **propionic** C3, **butyric** C4 (butter), **valeric** C5, **caproic / caprylic / capric** C6 / C8 / C10 (goat), **lauric** C12 (coconut oil), myristic C14, palmitic C16. **Fatty acids are carboxylic acids.**

Her naming exercise: megatomoic acid

14-carbon chain from the COOH, two C=C at C3 and C5: **tetradeca-3,5-dienoic acid**. The **-a- stays** in "tetradeca" (next letter is a consonant, d).

Note: -dienoic = two C=C; **-dioic** = two COOH groups. Don't confuse them.

PHYSICAL PROPERTIES

- Very polar, O–H H-bonds, and they form **dimers** (two molecules held by two H bonds act like one bigger molecule) → the **highest boiling and melting points** of the organic families.
- **Boiling point, high to low: carboxylic acid > alcohol > amine > aldehyde/ketone > ether > thiol > hydrocarbon.**
- **Solubility:** butanoic acid is infinitely soluble; **C6 slightly soluble, C7+ insoluble**.

Slide exercise: arrange by increasing boiling point

Aldehyde < alcohol < carboxylic acid (octanal 171 °C, 1-octanol 195 °C, octanoic acid 223 °C).

ACIDITY

- **Weak acids** (far weaker than HCl, H_2SO_4, HNO_3). **Methanoic > ethanoic > propanoic.**
- The **C=O pulls electrons** through the carbon and helps the O–H hydrogen leave as H^+ → **carboxylate ($RCOO^-$)**. Alcohol: one pull, not enough (except phenols). Aldehyde: H on carbon, weakest.
- **Inductive effect:** an extra electronegative atom or **EWG** near the COOH makes the acid **STRONGER**: CH_3COOH (pK_a 4.75) < $ClCH_2COOH$ (2.86) < $Cl_2CHCOOH$ (1.48) < **Cl_3CCOOH (TCA, 0.70)**.
- Form in water: **RCOOH at pH ≤ 2, 50:50 at pH = pK_a, $RCOO^-$ at pH ≥ 7**.

Her two exercises

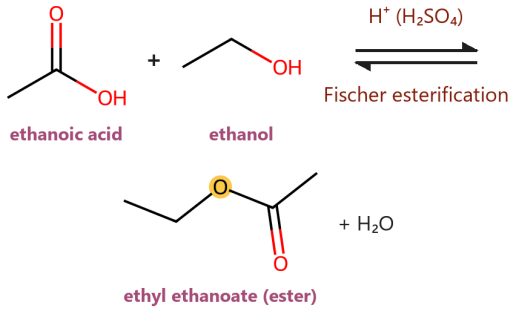
4-Nitrobenzoic acid is stronger than benzoic acid ($-NO_2$ is an EWG). **Lactic acid** (2-hydroxypropanoic) is stronger than propanoic acid (α-OH withdraws).

Note: –OH on a benzene ring DONATES (resonance, EDG, o/p director), but **–OH on a saturated carbon WITHDRAWS** (no resonance). This is the difference from the benzene chapter.

REACTIONS OF CARBOXYLIC ACIDS

- **1. Neutralization:** acid + base → **salt + H_2O**. Benzoic acid + NaOH → **sodium benzoate** (food preservative); with NH_3 → ammonium benzoate. **-oic acid → -oate**: butanoic acid + NaOH → **sodium butanoate** + H_2O. With $NaHCO_3$: salt + H_2O + **CO_2 gas**.
- **2. Fischer esterification:** acid + alcohol (**conc. H_2SO_4**) ⇌ **ester + H_2O**.

The OH of the ACID and the H of the ALCOHOL leave as water. The alcohol's O (highlighted) ends up between the C=O carbon and the alkyl group. The C=O stays (substitution).

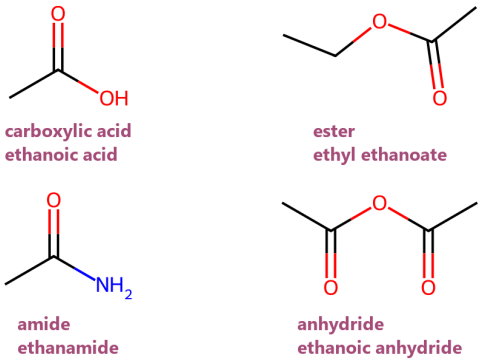


The OH of the acid and the H of the alcohol leave as water

- **Which atoms leave:** the **OH from the ACID** and the **H from the ALCOHOL** (not the other way around). The alcohol's O ends up between the carbonyl carbon and the alkyl group.
- Unlike hemiacetal formation, the **π bond stays**: **substitution** (OH → OR), not addition. **Reversible** (reverse = ester hydrolysis).
- Her examples: ethanoic acid + ethanol → **ethyl ethanoate**; with propan-2-ol → **isopropyl ethanoate** (O stays on the middle carbon). **Diacid + excess ethanol → diester.**
- **3. Decarboxylation:** loss of **CO_2**. Needs very high temperature without enzymes; **enzymes do it at body temperature**. **3-Oxobutanoic acid → acetone + CO_2.** Part of the CO_2 we exhale comes from the Krebs cycle.

DERIVATIVES: ANHYDRIDE, ESTER, AMIDE

- Mnemonic: **"see anhydride, see ester, see amide."**



The acid and its three derivatives

- **Anhydride** = two carboxylic acids minus water: **O between two carbonyl carbons**. **Ester** = carbonyl + **alkoxy (–OR)**. **Amide** = carbonyl + **nitrogen**. Cyclic amide = **β-lactam** (penicillin-type antibiotics).
- All are **derivatives** because **hydrolysis gives back the carboxylic acid**.

NAMING ESTERS

- **Two words, with a space, no dash:** alcohol part **-yl**, acid part **-oate**: **ethyl ethanoate**, **isopropyl ethanoate**, **methyl butanoate**.
- Number each part **separately from its own end** (acid part from the C=O carbon; alcohol part from the carbon on the O): **methyl 2-methylpropanoate**, **ethyl 2-hydroxypropanoate**.
- Ring on the carbonyl carbon: **carboxylate**: **methyl cyclohexanecarboxylate**. **Esters smell sweet** (fruits, perfume); thiols smell bad.

NAMING AMIDES AND ANHYDRIDES

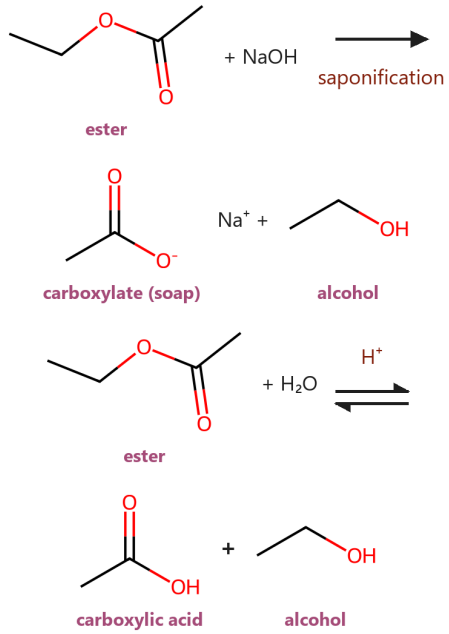
- **Amide:** C=O carbon is C1; drop -oic acid, add **-amide**: ethanamide, **2-methylpropanamide**, **3-hydroxypropanamide**. Ketone in the chain = **oxo-** (**2-oxobutanamide**). Groups on N use **capital N-**: **N-ethyl-N-methylethanamide**. Ring: **cyclohexanecarboxamide**.
- **Never locate the carbonyl in acids, esters, amides** (always C1). **The only carbonyl you locate is a ketone.**
- **Anhydride, symmetrical:** acid name with "acid" → **anhydride**: **ethanoic anhydride**. Different acids give a mixed anhydride (her board: ethyl propyl anhydride).

Ring suffixes (aldehyde / ester / amide)

carbaldehyde / carboxylate / carboxamide.

HYDROLYSIS (HYDRO = WATER, LYSIS = BREAKDOWN)

Anhydrides, esters and amides all give back the carboxylic acid (or its salt) on hydrolysis (anhydride + H_2O gives two acids).



Ester hydrolysis in base and in acid

- **Anhydride + H_2O → two carboxylic acids** (acetic anhydride is not "soluble" in water: it reacts).
- **Ester + NaOH → carboxylate salt + alcohol = saponification. Soaps are salts of carboxylic acids.** Ester + H_2O / H^+ → acid + alcohol (reverse of esterification).
- **Amide + H_2O (acid, heat) → carboxylic acid + ammonium salt.**

OTHER REACTIONS OF DERIVATIVES

- **Anhydride + NH_3 or amine → amide + a salt** (**2 moles** of amine: the second neutralizes the acid by-product).
- **Anhydride + alcohol → ester + carboxylic acid. Aspirin** = salicylic acid (its OH) + acetic anhydride → **acetylsalicylic acid** + acetic acid. Salicylic acid (willow bark) irritates the stomach; acetylating the OH makes it less acidic.
- **Ester + amine → amide + alcohol. Amides do NOT react with ammonia or amines.**

STEP-GROWTH POLYMERS

- Two monomers, **each with two functional groups**. **Polyamides** (diacid + diamine): **proteins** (peptide bond = amide), **Nylon-66**, **Kevlar**. **Polyesters** (diacid + diol): **Dacron**, **Mylar**. **Polycarbonates: Lexan** (bisphenol A disodium salt + phosgene; eyeglass lenses).

END OF LESSON

Topic 16: Introduction to Biochemistry

Unit XV: biomolecules, bonds, metabolism

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What to expect

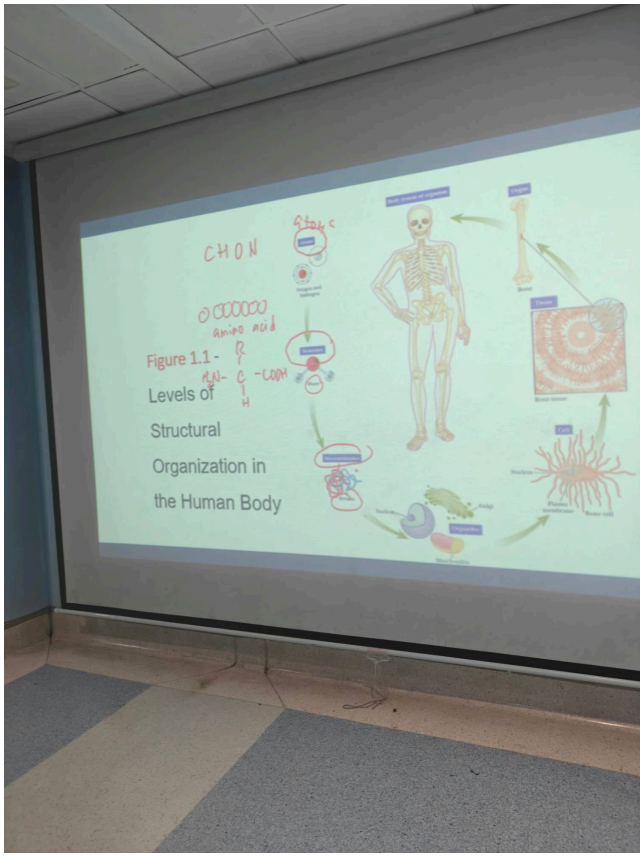
Slide exercises she used: **biochemistry covers carbohydrates, lipids, proteins and nucleic acids**; **anabolic vs catabolic** (protein synthesis and triacylglycerol formation = anabolic; polysaccharide hydrolysis = catabolic); **ATP** is the molecule that produces energy.

WHAT BIOCHEMISTRY IS

- The study of the **chemical substances in living organisms** and their **chemical interactions**: production, maintenance and reproduction of life.
- Early cells and biomolecules are thought to come from simple molecules: **H_2O, CH_4, NH_3, N_2, H_2**.
- **Bioinorganic: water (about 70% of body mass)** and inorganic salts. **Bioorganic (contain carbon): proteins, nucleic acids, lipids, carbohydrates** = the four **biomolecules**.
- **Medicine links:** nucleic acids → genetic diseases; proteins (sequence) → **sickle cell anemia**; lipids → **high LDL** → atherosclerosis; carbohydrates → **high blood glucose** → hyperglycemia, diabetes mellitus.

LEVELS OF STRUCTURAL ORGANIZATION

- **Atoms → molecules → macromolecules → organelles → cells → tissues → organs → body system / organism.** The four most common elements: **C, H, O, N ("CHON")**.
- **Molecules:** atoms share electrons (covalent). **Macromolecule:** amino acids joined = polypeptide = protein. **Mitochondria** = powerhouse; **nucleus** = control center. **Cell** = basic unit of life.



Her board work on Figure 1.1

FUNCTIONS AND FUNCTIONAL GROUPS

- **Lipids:** long-term energy storage, **membranes**. **Nucleic acids:** store and transmit genetic information. **Carbohydrates:** energy storage. **Proteins:** transport, structure, defense, **catalysis**.
- **Carbon is the backbone** of every biomolecule. **Bioenergetics** = energy flow in cells.
- Functional groups: carbohydrates **–OH**; fatty acids **carbonyl** (carboxyl); amino acids **–NH_2 and –COOH**; DNA **phosphate**; **ATP has three phosphates**.

MONOMERS AND POLYMERS

- **Amino acid → protein. Sugar (monosaccharide) → carbohydrate (polysaccharide). Nucleotide → nucleic acid.** Glucose + glucose → disaccharide → oligosaccharide → polysaccharide (**starch**).

Note: Lipids are NOT macromolecules (no common structure, no polymers). Only proteins, nucleic acids and carbohydrates are.

- **Informational macromolecules = proteins and nucleic acids** (the sequence of building blocks is not repeated). A polysaccharide repeats the same glucose, so it is **not** informational.

THE THREE BONDS (ALL BY DEHYDRATION SYNTHESIS)

- **Dehydration synthesis = condensation:** a water molecule is removed (–OH from one unit, –H from the other). Each bond has a **direction ("sense")**.

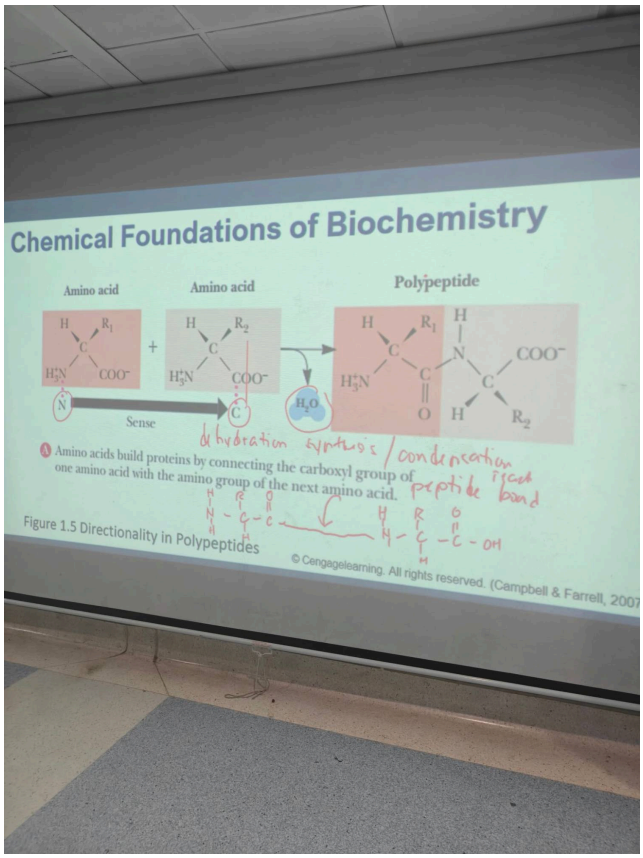
THREE BONDS, ALL BY DEHYDRATION SYNTHESIS

Biomolecule	Bond	Direction / link
Protein	peptide bond (C–N)	N-terminal → C-terminal
Carbohydrate	glycosidic bond	C1 of one glucose to C4 of the next (1→4)
Nucleic acid	phosphodiester bond	5' → 3' (3'–OH to the next 5' phosphate)

Each time, water is removed (–OH from one unit, –H from the other): also called a condensation reaction.

Peptide, glycosidic, phosphodiester

- **Peptide bond:** carboxyl of one amino acid + amino of the next; **N-terminal → C-terminal**.
- **Glycosidic bond: C1 of one glucose to C4 of the next (1→4)**.
- **Phosphodiester bond:** a nucleotide = **base + pentose sugar + phosphate**; the **3'-OH** links the next nucleotide; direction **5' → 3'**.



Her board work on Figure 1.5: N → C, H_2O removed, peptide bond

METABOLISM

- **Metabolism** = the sum of all life-sustaining chemical reactions. In 40 years the body processes about **6 tons of food** and **10,000 gallons of water**.
- **Metabolic pathway** = sequential reactions (product of one step = reactant of the next). **Metabolites** = the species made along a pathway. **Enzymes** (biological catalysts) are controlled by **upregulating, downregulating, inhibiting or activating**.
- **Anabolism: builds up** (small → large), **requires energy (ATP)**: protein synthesis, triacylglycerol synthesis, **photosynthesis** ($CO_2 + H_2O \rightarrow$ glucose $+ O_2$).
- **Catabolism: breaks down** (large → small), **releases energy (ATP)**, heat and waste: carbohydrate digestion, **hydrolysis**. Fully catabolized macromolecules end as **CO_2 and H_2O**.
- **ATP ⇌ ADP:** energy used: ATP (**tri**phosphate) → **ADP (diphosphate)**; energy absorbed: ADP → ATP.

Slide exercise: anabolic or catabolic?

Protein from amino acids = anabolic. Triacylglycerol from glycerol and fatty acids = anabolic. Hydrolysis of a polysaccharide = catabolic ("hydrolysis = break").

SITES AND TYPES OF PATHWAYS

- **Anaerobic:** no oxygen, in the **cytoplasm**. **Aerobic:** needs oxygen, in the **mitochondria**; O_2 is the **final electron acceptor**.

TYPES OF METABOLIC PATHWAYS

Type	Idea	Her example
Linear	series not repeated	glycolysis (glucose → pyruvate, ATP released)
Circular	final product is also an initial reactant	citric acid (Krebs) cycle: oxaloacetate + acetyl-CoA
Spiral	repeated reactions break down (or build up) a compound	β-oxidation: 8 C → 6 → 4 → 2 (3 rounds): 4 acetyl-CoA, 3 $FADH_2$, 3 NADH

Aerobic: needs O_2, in the mitochondria. Anaerobic: no O_2, in the cytoplasm.

Linear, circular, spiral

- β-oxidation: the target is the **β-carbon**; $FADH_2$ and NADH can be converted to ATP.

The goal of metabolism

Convert the **chemical energy in food** into chemical energy as **ATP**. The molecule responsible for producing energy: **ATP** (not AMP, ADP or NADH).

END OF LESSON

Topic 17: Proteins and Optical Isomerism

Unit XVI: amino acids, chirality, protein structure

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What she said she will NOT ask

No R/S assignment and **no determining levo/dextro** from a structure. R/S is a paper (IUPAC) label; d/l is measured with a polarimeter. Slide exercises: **classify side chains**, **name L/D from a Fischer projection**, **which tertiary force**, **which levels of structure** a protein has.

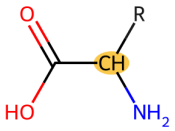
PROTEINS

- **Naturally occurring, unbranched polymer** of **amino acids**. **Most abundant substance in cells after water (about 15% of cell mass).**
- Elements **C, H, N, O, S**; average **nitrogen 15.4%**. The body **cannot store nitrogen**, so excess leaves as **urea, uric acid, ammonia**. Fe, P and metals appear only in specialized proteins.

AMINO ACIDS

- Both **–NH_2 and –COOH** on the **α-carbon**.

Highlighted: the α-carbon, which carries the amino group, the carboxyl group, H and the side chain R. The 20 standard amino acids differ only in R.



general α-amino acid
$H_2N–CH(R)–COOH$

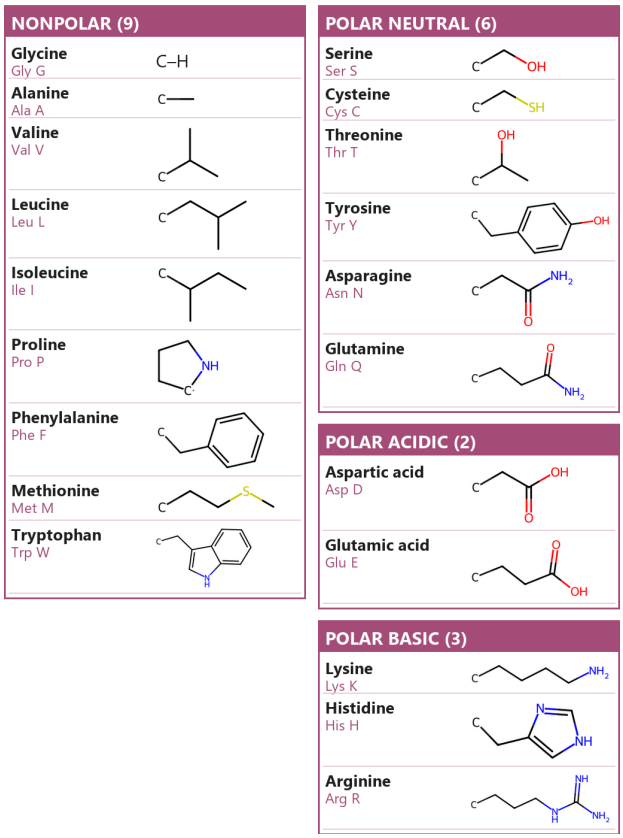
General α-amino acid

- **20 standard amino acids** differ in **R** (size, charge, polarity). **More than 700** amino acids are known; hydroxyproline is not one of the 20.
- **Essential** amino acids must come from the **diet**; nonessential ones the body makes. **A codon (3 nucleotides) codes for one amino acid.**

FOUR CLASSES BY SIDE-CHAIN POLARITY

- **Look ONLY at the side chain (R).**

THE 20 STANDARD AMINO ACIDS: classified by the SIDE CHAIN only
C at the left of each side chain = the α-carbon (the rest, $H_2N–CH–COOH$, is the same for all)



The 20 standard amino acids by class

- **Nonpolar (9):** hydrocarbon side chains; hide **inside** the folded protein. **Met** has S, but S and C are both **2.5**, so C–S–C is nonpolar. **Trp** has an N, but its large ring overpowers it.
- **Polar neutral (6): Ser, Thr, Tyr** have **–OH**; **Cys** has **–SH**; **Asn, Gln** have an amide (C=O and N). More soluble in water.
- **Polar acidic (2):** side-chain **–COOH** (names end in "acid"). **Polar basic (3):** side-chain **N** (amine).

Note: Glycine is polar as a whole molecule, but its side chain (H) puts it in the **nonpolar** class.

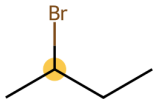
Slide exercise: classify R

CH_3 = nonpolar; CH_2OH = polar neutral; CH_2COOH = polar acidic; CH_2–ring = nonpolar; $(CH_2)_4NH_2$ = polar basic.

OPTICAL ISOMERISM

- **Optical isomers** = stereoisomers: **same formula and connectivity, different 3-D arrangement**, **non-superimposable mirror images**. **Chiral** = not superimposable (your **hand**); **achiral** = superimposable (a chair).
- A **chiral center** = a carbon with **four different groups** (four single bonds, no double bond; a ring carbon can qualify). **At least one chiral carbon = chiral molecule.**

C2 (highlighted) holds four different groups: H, Br, CH_3 and CH_2CH_3 (ethyl) = chiral center. C1, C3, C4 each have two or three identical groups (H), so they are not chiral.



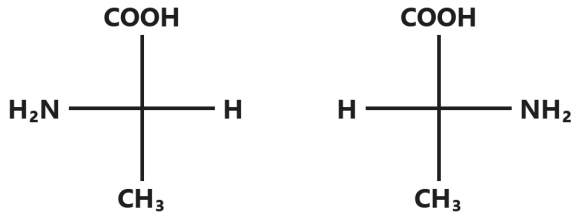
2-bromobutane

Her example: 2-bromobutane

- **Dextrorotatory (+, d)** rotates plane-polarized light **clockwise (right)**; **levorotatory (-, l)** counterclockwise (left). Measured with a **polarimeter**.
- Her examples: **dextrose = D-glucose** (the body accepts only D-glucose); **methamphetamine** d = "meth", l = Vicks inhaler; **lactic acid** (+) from muscle, (-) from milk; **limonene** from lemons vs oranges. Enzymes are specific (**lock-and-key, induced fit**).
- **n chiral centers → up to 2^n stereoisomers** (1 center = 2; 3 centers = 8).
- **Enantiomers:** non-superimposable mirror images; **same physical properties**, rotate light in **opposite directions**. **Diastereomers:** stereoisomers that are **not mirror images**; need **2+ chiral carbons**; **different physical and chemical properties**.

CHIRALITY OF AMINO ACIDS

- All standard amino acids have four different groups on the α-carbon **except glycine** (R = H) → **glycine is the only achiral standard amino acid.**
- Natural amino acids and proteins are **L** ("L for life"); **sugars are D**.



L-alanine
NH_2 on the LEFT

D-alanine
NH_2 on the RIGHT

Rule: –COOH on top, R at the bottom, –NH_2 on the horizontal. Left = L, right = D.

Fischer projection of alanine

- **Fischer rules: COOH on top, R at the bottom, NH_2 on the horizontal: left = L, right = D.**

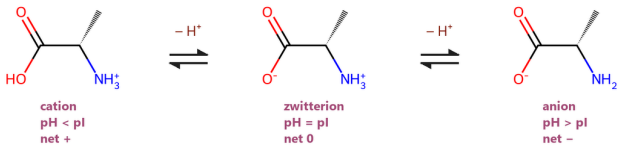
Slide exercise

Name each amino acid from the side its NH_2 is on: **L-isoleucine, D-cysteine, L-tyrosine.**

ACID-BASE PROPERTIES

- **Amphoteric:** –COOH (acid) and –NH_2 (base) on the same carbon. In neutral solution **COOH** → **COO^-** and **NH_2** → **NH_3^+**: no amino acid is neutral in the body.
- **Zwitterion:** + on one atom and - on another, **no net charge**.

Alanine at three pH values. Left of each arrow = more H^+ (acid). The zwitterion has a + and a – charge but no net charge.



cation
pH < pI
net +

zwitterion
pH = pI
net 0

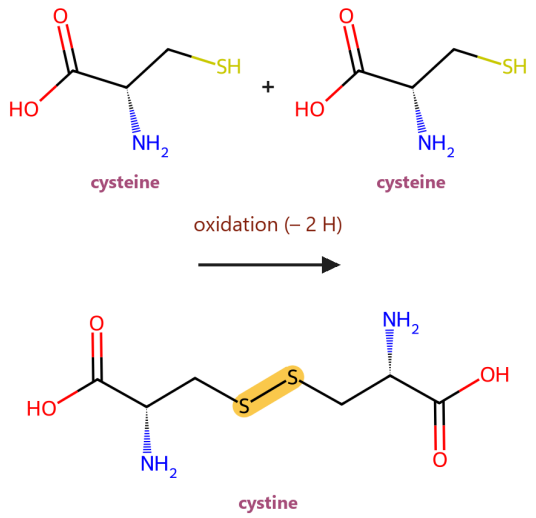
anion
pH > pI
net –

Cation, zwitterion, anion

- **Isoelectric point (pI):** the pH where the **zwitterion** predominates. The 15 nonpolar and polar neutral amino acids: **pI 4.8–6.3**; the **3 basic** have a **higher** pI, the **2 acidic** a **lower** one.
- **pH < pI:** net **positive**. **pH > pI:** net **negative.** Used in **gel electrophoresis** (also DNA fingerprinting). Different body regions have different pH, so **enzymes have different working pH**.

CYSTEINE AND PEPTIDES

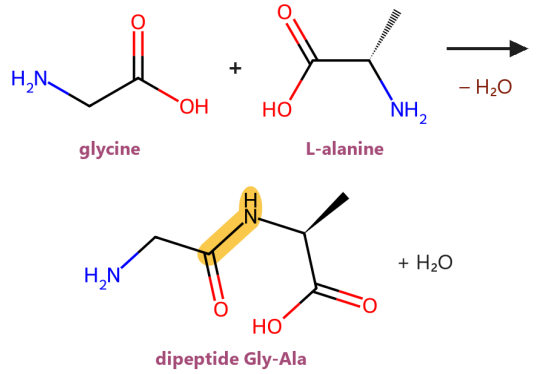
Two sulfhydryl (–SH) groups join in a covalent disulfide bond (–S–S–, highlighted): the strongest tertiary-structure bond.



Two cysteines → cystine

- **Cysteine** has a sulfhydryl (**–SH, thiol**); two cysteines **oxidize** to **cystine** with a covalent **disulfide bond (–S–S–)**.

Carboxyl of one amino acid + amino group of the next: OH and H leave as water. The new C–N bond (highlighted) is the peptide bond (an amide). Read N-terminal → C-terminal. n amino acids = n − 1 peptide bonds.



Peptide bond between two amino acids

- **Peptide bond** (covalent) = an **amide**, between the **amino** group of one and the **carboxyl** of another. **Dipeptide = 2 amino acids, 1 bond; tetrapeptide = 3 bonds (n - 1). Oligopeptide** 10-20 residues; **polypeptide** = long chain. Every peptide has an **N-terminal** and a **C-terminal** end.

FOUR LEVELS OF PROTEIN STRUCTURE

- **Primary:** the **sequence** of amino acids (peptide bonds are the backbone). Specify it by **the number and sequence of amino acids**. One wrong amino acid can cause disease: **sickle cell anemia = glutamate → valine at position 6 of the β-globin chain**; genetic. The six atoms of the peptide linkage lie in **one plane**.
- **Secondary:** the **backbone** arrangement: **α-helix** and **β-pleated sheet**, held by **H bonds between C=O and N–H of the backbone** (not R groups).
- **α-helix:** coiled spring, **right-handed**; H bonds **parallel to the axis**, each C=O to the N–H **4 residues** along; **3.6 residues per turn**; R groups **outward**; in **keratin**.
- **β-pleated sheet:** extended segments side by side, **intrachain** (U-turns) or **interchain**; H bonds **in the plane**; R groups **above and below**. Both occur only where R groups are **small**.
- **Tertiary:** the overall **3-D shape** from **side chain with side chain** interactions.
- **1. Disulfide bonds** (Cys–Cys): covalent, **the strongest**. **2. Salt bridges** (electrostatic): **acidic R ($–COO^-$) with basic R ($–NH_3^+$)**. **3. Hydrogen bonds:** polar R groups; weak, disrupted by pH changes. **4. Hydrophobic interactions:** nonpolar side chains (London dispersion).

Slide exercise

Glycine and asparagine: hydrogen bonding. Glutamic acid and lysine: electrostatic attraction (salt bridge).

- **Denaturation: alcohol** disrupts the **H bonds** in bacterial proteins. **Hair** (mostly nonpolar, held by disulfide bonds) is not dissolved by alcohol.
- **All proteins have up to tertiary structure; not all have quaternary.**
- **Quaternary:** only with **more than one polypeptide chain** (each a **subunit**); held **noncovalently** (electrostatic, H bonds, hydrophobic). **Allosteric:** a change in one subunit changes another. Possible for **multimeric**, not monomeric, proteins.

Hemoglobin vs myoglobin

Hemoglobin: tetramer **$\alpha_2\beta_2$** (α 141, β 146 residues), heme with **iron**, **4 O_2**, carries O_2 from the lungs; **quaternary**. **Myoglobin:** one chain, **1 O_2**, O_2 storage in muscle, **no quaternary**, **higher O_2 affinity**.

CLASSIFYING PROTEINS

- **Protein** = a peptide with **at least 40** residues. **By chains: monomeric** vs **multimeric**. **By composition: simple** (amino acids only) vs **conjugated** (+ a non-amino-acid **prosthetic group**).

SUMMARY TABLE: CONJUGATED PROTEINS

Class	Prosthetic group	Example (function)
Hemoproteins	heme	hemoglobin (O_2 carrier in blood), myoglobin (O_2 binder in muscle)
Lipoproteins	lipid	LDL, HDL (lipid carriers)
Glycoproteins	carbohydrate	gamma globulin (antibody), mucin (lubricant), interferon (antiviral)
Phosphoproteins	phosphate	glycogen phosphorylase (enzyme)
Nucleoproteins	nucleic acid	ribosomes (protein synthesis), viruses
Metalloproteins	metal ion	ferritin (iron storage), alcohol dehydrogenase (zinc)

> **Note:** She corrected herself: **myoglobin is conjugated**, not simple (heme with iron).

- **By shape: fibrous** = elongated, linear chains, **structural**. **Globular** = spherical, hydrophilic outside and hydrophobic inside. **Enzymes are globular** (they must be flexible).

END OF LESSON

Topic 18: Enzymes

Unit XVII: characteristics, classification, mode of action, specificity, factors, cofactors, inhibitors

CHEMISTRY (PROF. L. RANAY) | FINALS EVAL, NOV 14: TOPICS 15-18 (UNITS XIV-XVII)

What to expect

She said: **"I will not ask you to name" enzymes**, but **remember the classification** (the six classes and how to spot each). Slide exercises: **classify a reaction by its EC class** (a C–OH becoming C=O = oxidoreductase; water added to break a molecule = hydrolase).

WHY ENZYMES MATTER

- Catalysis is **the most important function of proteins**; enzyme-controlled reactions are **10^8 to 10^{11} times faster** than the same reactions without enzymes. Without them, digesting one meal would take years: **no continuity of life**.
- **Diagnosis:** in a **heart attack (myocardial infarction)**, blood **LDH** and **CK** rise; **CK shows up more rapidly than LDH**. **Therapy:** e.g., **digestive enzymes**.
- A reaction needs **1. enough activation energy** and **2. proper orientation** (collision theory). The enzyme handles **both**.

DEFINITION AND CHARACTERISTICS

- **Enzymes are biological catalysts** (**proteins**) that catalyze nearly all cell reactions by finding an **alternative pathway with a lower activation energy**.
- **Not altered or consumed**, so **reusable**; **highly specific** (unlike inorganic catalysts); **sensitive to their environment** (temperature, pH, substrate concentration).
- **Globular** shape with a complex 3-D structure (α-helices, β-sheets, side chains), so they travel easily in the aqueous body.

SIMPLE VS CONJUGATED ENZYMES

- **Simple:** protein only (few enzymes). **Conjugated:** protein + non-protein part (**almost all** enzymes).
- **Apoenzyme** = protein part alone, **inactive**. **Cofactor** = non-protein part. **Apoenzyme + cofactor = holoenzyme** (**active**).
- **Prosthetic group: tightly bound** (covalent bond or salt bridge); usually a small **inorganic** atom or molecule (**Fe**), may be organic (**FAD**); often stays intact in the reaction.
- **Coenzyme: loosely bound** (only intermolecular forces); **organic**, often **vitamin-derived** (**CoA, NADH**). **Coenzymes are a type of cofactor.**

> **Note: Vitamins and minerals** are needed because they are **cofactors** of our enzymes. Inactive forms often end in **-gen**, active forms in **-in**: **pepsinogen** (stomach) + **HCl (H^+)** → **pepsin**.

KEY TERMS AND NAMING

- **Active site:** the small region where the **substrate** attaches and the reaction happens. **Substrate:** the **reactant**; the enzyme is only the catalyst. The **primary structure** must fold the right residues into the active site.
- Names usually end in **-ase** (**sucrase** hydrolyzes sucrose); digestive enzymes keep common names (**pepsin, trypsin**); a name can show the function (**oxidase**) or substrate + function (**alcohol dehydrogenase**).

CLASSIFICATION (EC NUMBERS)

- By the **IUB**, by the **type of reaction**; each enzyme has an **EC number** of **four numbers separated by periods**.

SUMMARY TABLE: ENZYME CLASSES (EC NUMBERS)

EC	Class	Reaction catalyzed	Slide examples
1	**Oxidoreductases**	Oxidation-reduction reactions	glucose oxidase, peroxidase, catalase, phenylalanine hydroxylase
2	**Transferases**	Group transfer reactions	transaminases (ALT and AST), kinases (phosphotransferases), transmethylases, transpeptidases, transacylases
3	**Hydrolases**	Hydrolysis reactions; **water is the acceptor** of the transferred group	peptidases, amylase, maltase, lactase, lipase, deaminases, phosphatases
4	**Lyases**	Cleave bonds by means **other than hydrolysis and oxidation**; add or remove H_2O, NH_3 or CO_2 across double bonds	fumarase, carbonic anhydrase, hydratase, dehydratase
5	**Isomerases**	Isomerization reactions: L to D, mutase reactions (shifts of chemical groups)	isomerase, mutase
6	**Ligases**	**Join** two molecules with covalent bonds, using energy from **ATP**	acetyl-CoA carboxylase, glutamine synthetase

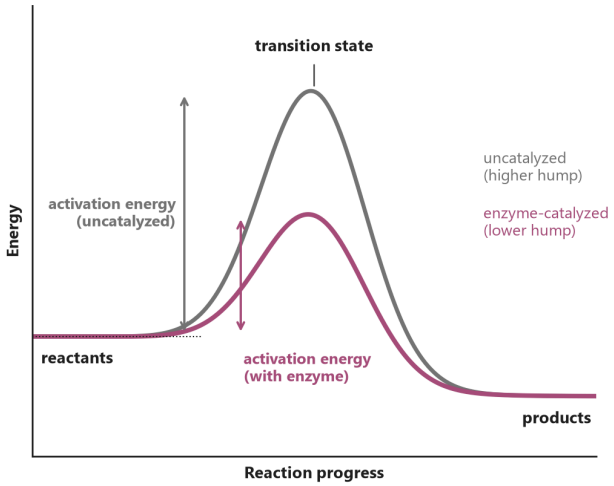
- **EC 7, translocases** (slide 18): move ions or molecules **across membranes**.
- **Reading EC 2.7.1.1 (hexokinase): 2** class (transferases); **7** subclass (transfer of a **phosphorus-containing group**); **1** sub-subclass (phosphotransferase, **alcohol group as acceptor**); **1** serial number. Systematic name **ATP:D-hexose-6-phosphotransferase**: glucose + ATP → **glucose-6-phosphate** + ADP.

HOW TO SPOT THE CLASS

- **Oxidoreductase:** OH → C=O (H removed), a new double bond. **Transferase:** a group moves (ATP → ADP + phosphorylated sugar). **Hydrolase: water** breaks a big molecule. **Lyase:** a bond cleaved **without** water or oxidation; a double bond appears or disappears. **Isomerase:** same atoms, only the **position** changes, or **L becomes D**. **Ligase:** two molecules joined, using **ATP**.

HOW ENZYMES WORK

- Every reaction needs the **activation energy**; at the top the molecules are in the **transition state**. Enzymes **lower** it by a new pathway ("a short cut"; going around the mountain, not removing it), with **no rise in temperature**, and they hold the reactants in the right **orientation**.

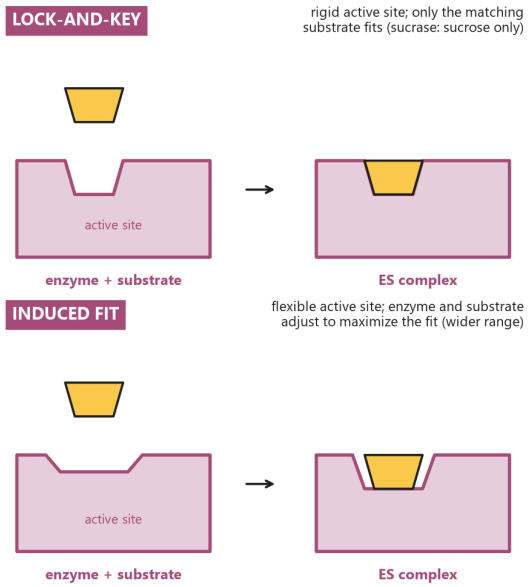


The enzyme gives a lower hump (lower activation energy); reactants and products stay at the same energy levels

- **Enzyme + substrate ⇌ enzyme-substrate (ES) complex → enzyme + product.** The ES complex is the **intermediate**.

LOCK-AND-KEY VS INDUCED FIT

- **Lock-and-key: rigid** active site; only the **matching shape** fits (substrate = key). **Sucrase** acts only on **sucrose**.
- **Induced fit: flexible** active site that changes shape; enzyme and substrate adjust to **maximize the fit**; a **greater range of substrate specificity**; fits **most enzymes**. **Alcohol dehydrogenase** accepts methanol, ethanol, isopropyl alcohol (all have the OH).



Lock-and-key (rigid site) vs induced fit (the site closes around the substrate)

SPECIFICITY

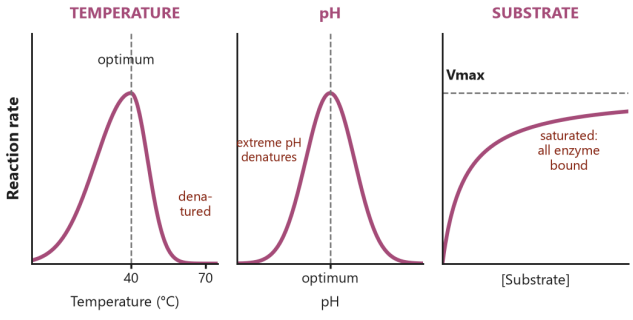
- **Starch** (α-glycosidic bonds) is digested, **cellulose** (**β**-glycosidic bonds) is not: we have no enzyme for the β bond. Our enzymes act on **D-sugars** and **L-amino acids** (an **L-glucose** IV would be treated as foreign).

SUMMARY TABLE: TYPES OF ENZYME SPECIFICITY

Type	What it means	Example
Absolute	One reaction for **one specific substrate**	**urease** hydrolyzes only urea; **sucrase** acts only on sucrose
Group	One reaction for substrates with **similar functional groups**	**hexokinase** adds phosphate to hexoses (glucose, fructose, mannose); **alcohol dehydrogenase** acts on a range of alcohols
Linkage	One type of reaction for a specific **type of bond**	**salivary amylase** hydrolyzes α-1,4-glycosidic bonds in starch; **pepsin** cuts peptide bonds next to aromatic or hydrophobic residues
Stereospecificity	Acts on one particular **stereoisomer** (chirality is inherent in the active site)	**L-amino acid oxidase** oxidizes L-amino acids but not D-amino acids

FACTORS AFFECTING ENZYME ACTIVITY

- **Temperature:** activity rises to the **optimum** (most enzymes about **40 °C**), then falls. **Cold-water fish die at 30 °C** (enzymes denature); a few **bacteria** withstand up to **100 °C**; most enzymes are **fully denatured at 70 °C**.
- **pH:** a slightly off pH changes the **charges** of enzyme and substrate (ionization), which affects **substrate binding**; each enzyme has its own **optimum pH**; **extreme pH** denatures.
- **Substrate concentration:** the rate rises until the enzyme is **saturated** (all enzymes bound to substrate) = **saturation point**, maximum activity (**Vmax**). Changing the **enzyme** concentration changes **Vmax**.
- **Enzyme concentration:** the rate **increases** with enzyme concentration (when enzyme is much lower than substrate).



Rate vs temperature, pH and substrate concentration

COFACTORS AND COENZYMES

- A cofactor is an extra **non-protein** molecule that **assists**. Tightly bound = **prosthetic group**; bound and released easily = **coenzyme**. **Many vitamins are coenzymes**, **not specific** to which enzyme they assist.

METAL COFACTORS

- **Cu^{2+}:** cytochrome oxidase (last enzyme of the electron transport chain). **Fe^{2+}/Fe^{3+}:** catalase (H_2O_2 → water + oxygen). **Mg^{2+}:** hexokinase (glucose → glucose-6-phosphate). **Mn^{2+}:** arginase (final enzyme of the urea cycle). **Zn^{2+}:** alcohol dehydrogenase, with NAD^+.

COENZYMES AND THEIR VITAMINS

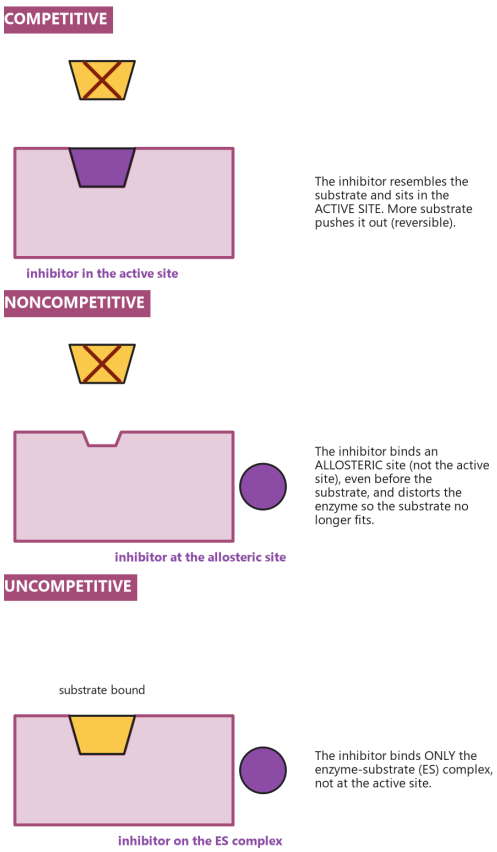
- **FAD/$FADH_2$ (B2):** reducing agent; donates electrons to the electron transport chain. **NAD/NADH (B3):** carries electrons; oxidizing agent. **NADH and $FADH_2$ are electron carriers** feeding the last part of glucose metabolism (**ATP**).
- **CoA (from B5): acyl group carrier**; acetyl-CoA starts the **Krebs cycle**. **Coenzyme Q10 (ubiquinone):** oil-soluble, carries electrons in aerobic respiration. **Quinones:** conjugated aromatic rings, pigments.
- **Vitamin C:** co-substrate with iron in **collagen** formation. **TPP (B1):** citric acid cycle, **carboxyl** group transfer. **B6: amino** group transfer. **Biotin (B7): carboxyl** transfer. **THF (B9): methylation**. **Cobalamin (B12): methyl** transfer. Also vitamins A and K.

INHIBITORS

- Substances that **reduce the rate**; usually **specific**, work at **low concentrations**; they **block** the enzyme but usually do not destroy it. Many **drugs and poisons** inhibit enzymes. **Reversible or irreversible.**

SUMMARY TABLE: TYPES OF ENZYME INHIBITION

Type	Binds where	Resembles the substrate?	Effect of more substrate	Reversible?
Competitive	The **active site** (competes with the substrate)	**Yes**, closely	**Reverses** the inhibition	Reversible
Noncompetitive	**Not the active site** (allosteric site); distorts the enzyme's shape	No (different structure)	**Not influenced** by substrate concentration	Reversibly or irreversibly
Uncompetitive	Only the **enzyme-substrate complex**, not at the active site	No (different structure)	Substrate **binds longer** in the enzyme and the rate drops	Slide says it binds irreversibly
Irreversible	Listed as a mode of action; not detailed on the slides	Not stated	Not stated	Irreversible



Where each inhibitor binds

- **Competitive:** the inhibitor enters the active site; **more substrate** pushes it out. **Methanol poisoning** is treated with **ethanol**, which competes with methanol for the active site of alcohol dehydrogenase.
- **Noncompetitive:** binds the **allosteric site**, changes the enzyme's shape, so the substrate no longer fits; acts **even before** the substrate binds. **Uncompetitive:** acts **only on the ES complex** (the substrate is already bound), also at an allosteric site.

APPLICATIONS OF INHIBITORS

- **Poisons:** snake bite, plant alkaloids, nerve gases. **Medicine:** antibiotics, sulfonamides, sedatives, stimulants.
- **Aspirin:** inhibits **prostaglandin** synthesis (aches of arthritis). **Methotrexate:** cancer chemotherapy, inhibits **DNA synthesis** of malignant cells. **Sulfa drugs:** inhibit **folic acid synthesis** that bacteria need.

ALLOSTERIC REGULATION AND FEEDBACK CONTROL

- **Allosteric** = binds a site other than the active site (not all enzymes are). **Negative regulator (inhibitor):** lowers the rate; the active site accepts substrate less readily. **Positive regulator (activator):** raises the rate. Example: **fructose-2,6-bisphosphate activates phosphofructokinase-1** (more glycolysis in response to **insulin**). Slide example: inhibitor **CTP**, activator **ATP**.
- **Feedback control:** a **product of a sequence** activates or inhibits the **first reaction** of that sequence (usually the final product inhibits the first step).

END OF LESSON